

Langgraph

Langchain : Langchain is like a giant toolbox + duct tape kit. It gives you abstractions (chains, agents, memory, tools, retrievers, etc.) that you can snap together to make an LLM-powered app.

Langgraph : Langgraph is more like a circuit board where you draw out the flow of your app. Instead of chaining functions in a linear sequence, it leans on graph-based workflows.

→ You define nodes (like "call LLM," "search database," "filter output") and edges (the paths between them). It's built on top of Langchain but brings structure for handling loops, branching, and state management in a cleaner way.

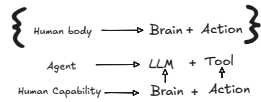
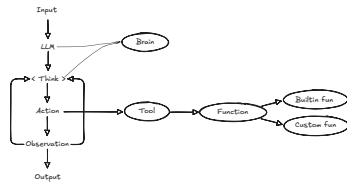
what is AI agent ??

AI agents are basically LLMs with a brain + arms.

The brain 🧠 is the large language model (like GPT), which can understand and generate text.

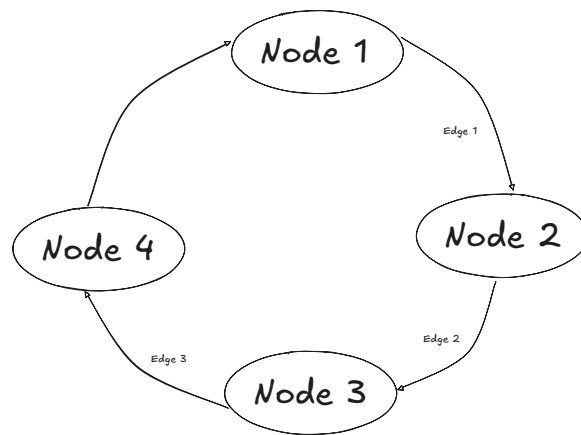
The arms 🦾 are the tools or actions it can take — like searching Google, calling an API, querying a database, or running code.

→ So instead of just chatting with you, an AI agent can act in the world. Example: you ask it to "look at the cheapest flight tomorrow," and it won't just explain how — it'll actually search, compare, and take steps to book.

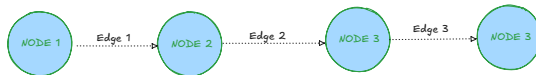


→ In LangGraph, an Orchestrator is a controller node that takes an input, decides how to break the work into smaller subtasks, sends those subtasks to other nodes (called workers), and then collects their results into a final output.

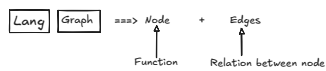
Cyclic Graph



Acyclic Graph



Graph ==> NODE + EDGE



Agentic Flow

